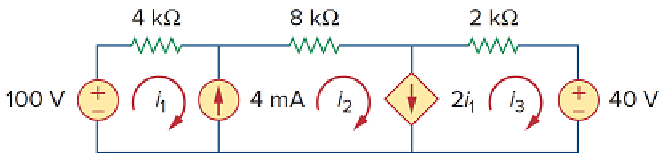
Chapter 3, Problem 62P

Problem

Find the mesh currents *i*1, *i*2, and *i*3 in the network of Fig. 3.106.

Figure 3.106:



Step-by-step solution

Step 1 of 4

Refer to Figure 3.106 in the textbook.

From the Figure 3.106,

$$i_2 - i_1 = 4 \text{ mA}$$

$$-i_1 + i_2 = 4\times10^{-3} \text{ ...... (1)}$$

Write the super mesh equation.

$$-100 + 4\text{k}i_1 + 8\text{k}i_2 + 2\text{k}i_3 + 40 = 0$$
$$4\text{k}i_1 + 8\text{k}i_2 + 2\text{k}i_3 = 60$$

$$2i_1 + 4i_2 + 1i_3 = 30\times10^{-3} \text{ ...... (2)}$$

From the Figure 3.106,

$$i_2 - i_3 = 2i_1$$

$$-2i_1 + i_2 - i_3 = 0 \text{ ...... (3)}$$

Step 2 of 4

Write the equations (1), (2) and (3) in matrix form.

$$\begin{bmatrix} -1 & 1 & 0 \\ 2 & 4 & 1 \\ -2 & 1 & -1 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 4\times10^{-3} \\ 30\times10^{-3} \\ 0 \end{bmatrix}$$

Calculate the determinants.

$$\Delta = \begin{vmatrix} -1 & 1 & 0 \\ 2 & 4 & 1 \\ -2 & 1 & -1 \end{vmatrix}$$
$$= -1(-4-1) - (1)(-2+2) + 0(2+8)$$
$$= 5$$

$$\Delta_1 = \begin{vmatrix} 4\times10^{-3} & 1 & 0 \\ 30\times10^{-3} & 4 & 1 \\ 0 & 1 & -1 \end{vmatrix}$$
$$= 4\times10^{-3}(-4-1) - (1)\left(-30\times10^{-3}\right) + 0\left(30\times10^{-3}\right)$$
$$= -20\times10^{-3} + 30\times10^{-3}$$
$$= 10\times10^{-3}$$

Step 3 of 4

Calculate the determinants.

$$\Delta_2 = \begin{vmatrix} -1 & 4\times10^{-3} & 0 \\ 2 & 30\times10^{-3} & 1 \\ -2 & 0 & -1 \end{vmatrix}$$
$$= -1\left(-30\times10^{-3}\right) - \left(4\times10^{-3}\right)\left(-2+2\right) + 0\left(0+60\times10^{-3}\right)$$
$$= 30\times10^{-3}$$

$$\Delta_3 = \begin{vmatrix} -1 & 1 & 4\times10^{-3} \\ 2 & 4 & 30\times10^{-3} \\ -2 & 1 & 0 \end{vmatrix}$$
$$= -1\left(0-30\times10^{-3}\right) + (-1)\left(0+60\times10^{-3}\right) + 4\times10^{-3}(2+8)$$
$$= 30\times10^{-3} - 60\times10^{-3} + 40\times10^{-3}$$
$$= 10\times10^{-3}$$

Step 4 of 4

Calculate the mesh currents using Cramer's rule.

$$i_1 = \frac{\Delta_1}{\Delta}$$
$$= \frac{10\times10^{-3}}{5}$$
$$= 2 \text{ mA}$$
$$i_2 = \frac{\Delta_2}{\Delta}$$
$$= \frac{30\times10^{-3}}{5}$$
$$= 6 \text{ mA}$$
$$i_3 = \frac{\Delta_2}{\Delta}$$
$$= \frac{10\times10^{-3}}{5}$$
$$= 2 \text{ mA}$$

Therefore, the mesh currents, i_1, i_2 and i_3 are $\boxed{2 \text{ mA, } 6 \text{ mA and } 2 \text{ mA}}$ respectively.